



• MECHANICAL ENGINEERING • LESSON 01

Introduction to Mechanical Engineering

Design, Build, and Optimise the Physical World

Lesson 1 of 30 | Mechanical Engineering Program | Beginner

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- The 5 core disciplines of mechanical engineering explained
- Engineering design process — from concept to manufactured product
- Fundamental concepts: forces, stress, strain, thermodynamics
- Essential software tools: SolidWorks, AutoCAD, ANSYS, MATLAB
- Mechanical engineering career pathways and salary ranges

01 | What Is Mechanical Engineering?

Mechanical engineering is one of the broadest and oldest engineering disciplines. It applies the principles of **physics, mathematics, and materials science** to design, analyse, manufacture, and maintain mechanical systems. From the engine in a car to the turbine in a power plant, from a prosthetic limb to a space launch vehicle — mechanical engineers are behind it all.

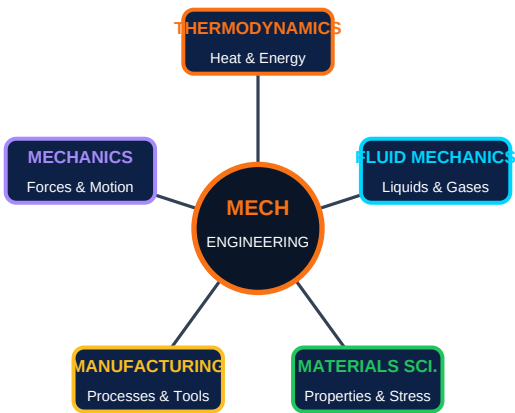
At its core, mechanical engineering asks: **"How do forces, energy, and motion interact — and how can we harness them to build useful things?"** It is both a creative discipline (inventing new machines and products) and an analytical one (calculating whether designs will survive in the real world).

SCOPE — What Mechanical Engineers Actually Build

Mechanical engineers design everything that moves or transfers energy: engines, turbines, pumps, compressors, robots, HVAC systems, aircraft structures, medical devices, manufacturing machines, vehicles, renewable energy systems, and consumer products. If it has a moving part or involves heat and energy — a mechanical engineer designed it.

02 | The 5 Core Disciplines of Mechanical Engineering

Mechanical engineering is built on five interconnected scientific disciplines. Every real-world mechanical design problem requires knowledge from multiple of these areas simultaneously.



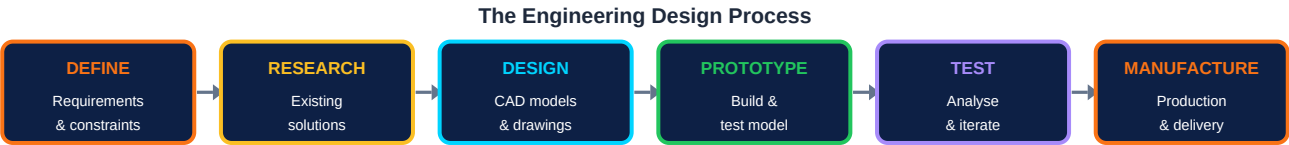
5 Core Disciplines of Mechanical Engineering

Discipline	Core Question	Key Concepts	Example Application
Thermodynamics	How does heat energy transfer and convert to work?	Temperature, pressure, entropy, efficiency, Carnot cycle, heat engines	Diesel engine, refrigerator, power plant turbine
Fluid Mechanics	How do liquids and gases flow and exert forces?	Pressure, viscosity, Reynolds number, Bernoulli, drag, lift	Pump design, aircraft wing, pipe networks, hydraulics
Materials Science	How do materials behave under load and temperature?	Stress, strain, yield strength, fatigue, hardness, corrosion	Choosing steel grade for a bridge, composites for an aircraft

Mechanics (Statics & Dynamics)	How do forces cause motion or keep structures in equilibrium?	Newton's laws, moments, free body diagrams, kinematics, vibration	Gear design, structural analysis, robot arm motion
Manufacturing	How are components designed to be efficiently made?	Tolerances, machining, welding, casting, CNC, 3D printing, GD&T;	Designing parts for production: minimise waste, maximise precision

03 | The Engineering Design Process

Mechanical engineering is not just about knowing equations — it is about using knowledge to **solve real problems through a structured design process**. This iterative cycle takes an idea from requirements through analysis, prototyping, testing, and into production. Professional engineers follow this process on every project, from a simple bracket to a complete machine.



Step	Key Activities	Deliverables	Tools Used
1. Define	Understand the problem: customer requirements, constraints, standards, safety requirements	Design brief, requirements specification, acceptance criteria	Meetings, standards documents (ISO, ASME, BS)
2. Research	Study existing solutions, patents, competitor products; identify failure modes to avoid	Literature review, benchmarking report, failure analysis	Google Scholar, patent databases, industry journals
3. Design	Generate concepts, select best option, create detailed CAD models and engineering drawings	3D CAD models, 2D engineering drawings, BOM (Bill of Materials)	SolidWorks, CATIA, Inventor, AutoCAD, Fusion 360
4. Prototype	Build a physical or digital prototype of the design to check form, fit, and function	Physical prototype, FEA simulation, CFD model	CNC machining, 3D printing, ANSYS, SolidWorks Simulation
5. Test	Test against requirements: strength, fatigue, thermal, dimensional, safety standards	Test reports, design changes, validation records	UTM (stress testing), thermal chambers, metrology CMM
6. Manufacture	Plan and execute production: tooling, quality control, assembly documentation	Manufacturing drawings, process sheets, QC plans	ERP systems, CNC programs, inspection gauges

04 | Essential Software Tools for Mechanical Engineers

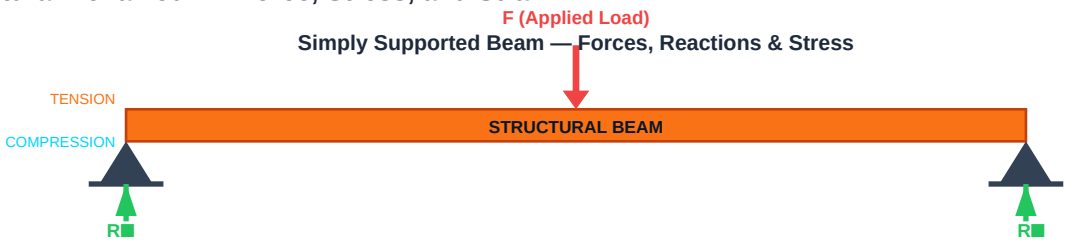
Software	Category	Primary Use	Industry Standard?
SolidWorks	3D CAD	Parametric 3D modelling, assemblies, drawings, simulation	YES — most widely used in industry

AutoCAD / AutoCAD Mechanical	2D CAD	Engineering drawings, technical drafting, GD&T; annotation	YES — standard for 2D drawings
CATIA / NX	3D CAD (Advanced)	Complex surface modelling, aerospace & automotive design	YES — aerospace, automotive OEMs
ANSYS	FEA / CFD Simulation	Structural, thermal, fluid, and modal analysis	YES — engineering simulation leader
SolidWorks Simulation	FEA (Integrated)	Stress analysis, fatigue, thermal within SolidWorks	YES — common for design validation
MATLAB / Simulink	Calculation & Control	Engineering calculations, dynamic system modelling, control systems	YES — widely used in research & R&D;
Fusion 360	3D CAD + CAM	Design + machining path generation for CNC	Growing — popular for SMEs and makers
SAP / Oracle ERP	Engineering Data	Bill of materials, procurement, production planning	YES — enterprise manufacturing

05 | Core Mechanical Concepts Every Engineer Must Know

Before using software, a mechanical engineer must understand the physical principles behind the calculations. These are the foundational concepts that underpin every structural, thermal, and dynamic analysis.

Structural Behaviour — Force, Stress, and Strain:



Concept	Formula	Units	Engineering Meaning
Force (F)	$F = m \times a$	Newton (N)	Any push or pull acting on a body
Stress (σ)	$\sigma = F / A$	Pa or MPa	Force per unit area — how hard material is being pushed/pulled
Strain (ϵ)	$\epsilon = \Delta L / L$	Dimensionless	Fractional deformation — how much it stretches relative to original
Young's Modulus (E)	$E = \sigma / \epsilon$	GPa	Material stiffness — resistance to elastic deformation
Factor of Safety (FoS)	$FoS = \sigma_{ult} / \sigma_{applied}$	Dimensionless	How much stronger the part is than the worst load — target ≥ 2.0 for structures
Pressure (P)	$P = F / A$	Pa or bar	Force per unit area in fluids — drives pumps, pipes, cylinders
Power (P)	$P = F \times v = T \times \omega$	Watt (W)	Rate of energy transfer — critical for motor and engine sizing

06 | Mechanical Engineering Career Pathways

Role	Core Responsibilities	Key Skills	Avg Salary (USD/yr)
Mechanical Design Engineer	Create 3D models and 2D drawings for components and assemblies	SolidWorks/CATIA, GD&T, tolerancing, material selection	\$60,000 – \$95,000
Structural Engineer	Analyse forces, stress, and fatigue in components and structures	FEA (ANSYS), statics/dynamics, material properties	\$65,000 – \$100,000
Thermofluids Engineer	Design HVAC, cooling, combustion, and fluid systems	CFD, thermodynamics, heat transfer, SolidWorks Flow	\$65,000 – \$105,000
Manufacturing Engineer	Optimise production processes, tooling design, quality control	CNC, GD&T, Lean/Six Sigma, ERP, tolerance analysis	\$60,000 – \$95,000
Maintenance Engineer	Keep industrial machinery running — preventive & corrective maintenance	Vibration analysis, hydraulics, pneumatics, failure diagnosis	\$50,000 – \$80,000
Project Manager (Mech)	Lead mechanical engineering projects from concept to delivery	MS Project, cost control, risk management, contracts	\$75,000 – \$120,000

Reflection:

"Pick any machine you interact with daily — a fan, a pump, a car door hinge, an elevator. Draw a simple free body diagram: what forces act on it? Where does energy enter and leave? What material must it be made from to survive those forces? You have just done the first step of mechanical engineering design."

Action checklist:

Next up — MECH-002:

Community & support:

Engineering Materials — metals, polymers, composites, ceramics;t.me/ayetechub
properties, selection criteria, and failure modes.